ADITYA UNIVERSITY

SCHOOL OF ENGINEERING

Department of Data Science

CERTIFICATE

This is to certify that the Cornerstone project work entitled *"***ONLINE SHOPPING CART SIMULATION***"* is submitted by

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in partial fulfillment of the requirements for the award of the B.Tech. degree in Data Science for the academic year 2025-2026.

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DECLARATION

We hereby declare that the Cornerstone project entitled "**ONLINE SHOPPING CART SIMULATION**" is original and is submitted to **ADITYA UNIVERSITY**, Surampalem, in partial fulfillment of the requirements for the award of the **B.Tech. degree** in **Data Science**. We further declare that this project work has not been submitted, either in full or in part, for the award of any degree in this or any other institution. We also confirm that this work complies with the Institutional Academic Integrity and AI Usage Policy

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ABSTRACT

The Online Shopping Cart Simulation is a desktop-based application developed to simulate the working of an e-commerce shopping system. The system allows users to view available products, add items to a cart, update quantities, remove products, and generate the final bill efficiently. The application is developed using Core Java, Java Swing for the graphical user interface, JDBC for database connectivity, and Oracle database for backend storage. The system ensures accurate price calculation, reduces manual effort, and provides a user-friendly interface for smooth shopping operations. This project helps in understanding real-time shopping cart functionality and database integration in modern applications.

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ACRONYMS

- **ONSCS** : Online Shopping Cart Simulation
- **JDBC** : Java Database Connectivity
- **GUI** : Graphical User Interface
- **SQL** : Structured Query Language

1. INTRODUCTION

1.1 Brief Information about the Project

The Online Shopping Cart Simulation is a desktop-based application designed to replicate the functionality of an online shopping system. It provides users with the ability to browse products, add them to a cart, update quantities, and generate a bill. The system replaces manual shopping processes with a digital solution, improving accuracy and efficiency. The application is developed using Java for backend logic, Swing for GUI design, JDBC for database connectivity, and Oracle for data storage.

1.2 Motivation and Contribution of Project

In many traditional shopping systems, product selection and billing are handled manually, which leads to inefficiencies such as calculation errors, difficulty in tracking selected items, lack of proper record maintenance, and delays during the checkout process. Customers often need to spend more time in billing queues, and manual calculations may result in incorrect totals. These limitations reduce the efficiency of the shopping process and affect user satisfaction.

To overcome these issues, the Online Shopping Cart Simulation has been developed as a digital solution that automates product selection, cart management, and billing operations. The system provides a structured and efficient way to handle shopping activities, ensuring accuracy, speed, and convenience. It allows users to interact with the system easily through a graphical interface and perform all shopping-related operations in a simplified manner.

The key contributions of this project include:

- Providing a digital platform for managing product selection and shopping activities in an organized manner
- Reducing manual effort by automating cart operations and billing processes
- Minimizing human errors in calculations and ensuring accurate bill generation
- Improving efficiency by enabling faster product selection and checkout
- Allowing users to easily add, update, and remove items from the shopping cart
- Enhancing user experience through an interactive and user-friendly graphical interface
- Supporting real-time calculation of total cost based on selected items and quantities

This system simplifies the overall shopping and billing process while improving accuracy, efficiency, and user convenience. It also provides a practical understanding of how modern online shopping systems function.

1.3 Objectives of the Project

The main objectives of the Online Shopping Cart Simulation are:

- To design and develop a user-friendly shopping cart system
- To simulate real-time product selection and purchasing
- To maintain product and cart data efficiently
- To implement automatic bill calculation
- To ensure smooth navigation between different modules
- To enhance user experience through simple interface design

1.4 Scope of the Project

The Online Shopping Cart Simulation is designed to provide a basic yet effective platform for simulating the core functionalities of an online shopping system. The scope of the project includes product display, selection of items, cart management, and billing operations. Users can browse available products, add items to the shopping cart, update quantities, remove unwanted items, and view the total cost before checkout. The system focuses on providing a simple and user-friendly interface to ensure ease of use and better interaction. It is mainly intended for educational purposes and can be used to understand the working principles of e-commerce applications. Although the system does not include advanced features such as online payment gateways, user authentication, or real-time order tracking, it serves as a strong foundation for developing more complex and scalable online shopping systems in the future. The project can be further extended by integrating additional functionalities to meet real-world requirements.

2. SYSTEM ANALYSIS

2.1 Functional Requirements

Functional requirements define the core operations that the Online Shopping Cart Simulation (ONSCS) must perform to ensure proper functioning of the system. These requirements describe what the system should do and how it interacts with users.

- **User Registration**
  The system should provide a registration feature that allows new users to create an account by entering details such as username, email, and password. It should ensure that all inputs are valid and prevent the creation of duplicate user accounts.
- **User Login**
  The system should provide a login feature where registered users can access their accounts using valid credentials. Authentication should be performed to ensure secure access.
- **Product Display**
  The system should display a list of available products along with details such as product name, price, and description. Users should be able to easily browse and select items.
- **Add to Cart**
  The system should allow users to add selected products to the shopping cart. Each selected item should be stored along with its quantity.
- **Update Cart Items**
  The system should provide functionality to update the quantity of items already added to the cart. Changes should be reflected immediately in the total price.
- **Remove Items from Cart**
  Users should be able to remove unwanted products from the shopping cart. A confirmation option can be included to avoid accidental deletion.
- **View Cart**
  The system should display all items added to the cart along with their quantities, individual prices, and total cost.
- **Billing and Total Calculation**
  The system should automatically calculate the total bill based on selected items and their quantities. It should ensure accurate and real-time calculation.

- **Search Product**

  The system may include a search feature to allow users to quickly find products by name or category.

2.2 Non-Functional Requirements

Non-functional requirements define the quality attributes and performance standards that the system must satisfy. These requirements focus on how the system performs rather than what it does.

- **Performance**

  The system should respond quickly to user actions such as adding items to the cart, updating quantities, and generating bills without noticeable delay.

- **Security**

  The system should ensure that user data and transaction details are protected. If login is implemented, proper authentication mechanisms should be used.

- **Usability**

  The system should have a simple, user-friendly interface that allows users to navigate easily without requiring technical knowledge.

- **Reliability**

  The system should function consistently without crashes or errors. It should handle unexpected inputs and maintain stable performance.

- **Scalability**

  The system should be capable of handling an increasing number of products and users without affecting performance.

- **Maintainability**

  The system should be easy to update and modify. Developers should be able to add new features or fix issues without major changes.

2.3 Requirements Specification

This section specifies the hardware and software needed to run the system.

2.3.1 Minimum Hardware Requirements

- Processor: Intel i3 or above
- RAM: 4 GB (minimum)
- Hard Disk: 20 GB free space
- Monitor: Standard display
- Keyboard and Mouse

2.3.2 Software Requirements

- **Operating System:** Windows 7/10/11
- **Programming Language:** Java
- **IDE:** VS Code
- **Database:** Oracle Database
- **Connectivity:** JDBC

3. TECHNOLOGY DESCRIPTION

3.1 Programming Language

3.1.1 Introduction to Java

Java is a high-level, object-oriented programming language developed by Sun Microsystems. It is widely used for developing desktop, web, and enterprise applications due to its platform independence and robustness.

In the Online Shopping Cart Simulation, Java is used to implement the core functionality of the application. It handles operations such as displaying products, adding items to the cart, updating quantities, removing items, and calculating the total bill. Java also manages the interaction between the user interface and the database, ensuring smooth execution of shopping activities.

Java provides important features such as object-oriented programming, exception handling, and strong memory management, which help in building reliable and efficient applications. These features make the system more organized, secure, and easy to maintain.

One of the main advantages of Java is that it is platform-independent, meaning the program can run on any system that has a Java Virtual Machine (JVM). This makes it suitable for developing portable applications like the Online Shopping Cart Simulation.

3.2 Framework / Libraries

3.2.1 Introduction to Swing

Swing is a part of Java that is used to create graphical user interfaces (GUI). It provides a set of components such as buttons, labels, text fields, tables, and frames to design user-friendly applications.

In this project, Swing is used to design the user interface of the Online Shopping Cart Simulation. It allows users to interact with the system easily through windows such as:

- Login Window
- Registration Window
- Product Display Window
- Shopping Cart Window
- Billing Window
- Order History window

Swing provides flexibility in designing layouts and improves the overall user experience. It makes the application visually interactive and easy to use.

3.3 Database Technology

The Online Shopping Cart Simulation uses a relational database for storing and managing product and user-related information in a structured manner. The database ensures efficient data storage, retrieval, and manipulation, which is essential for maintaining accuracy and consistency in the system. In this project, Oracle Database is used as the backend, and JDBC is used to connect the Java application to the database.

3.3.1 JDBC (Java Database Connectivity)

JDBC is an API (Application Programming Interface) used in Java to connect and interact with databases. It allows Java applications to execute SQL queries and retrieve or update data in the database.

In the Online Shopping Cart Simulation, JDBC is used to connect the Java application with the database (Oracle). It performs operations such as:

- Establishing a connection to the database
- Executing SQL queries (INSERT, UPDATE, DELETE, SELECT)
- Retrieving product and cart data from the database
- Updating cart details and product information
- Closing the database connection

JDBC acts as a bridge between the application and the database, ensuring smooth communication and data transfer. It helps in maintaining data consistency and enables efficient data handling.

3.3.2 Oracle Database

Oracle Database is a powerful relational database management system used for storing and managing structured data. It provides high performance, reliability, and security for handling large volumes of data.

In the Employee Management System, Oracle Database is used to store employee details such as employee ID, name, department, designation, and contact information. It supports SQL for data manipulation and ensures data integrity through constraints like primary keys and data types.

3.4 Tools & IDEs

3.4.1 Visual Studio Code

Visual Studio Code (VS Code) is used as the development environment for the Online Shopping Cart Simulation. It is a lightweight code editor that supports programming through various extensions. VS Code provides features such as syntax highlighting,

code completion, and debugging support, which help improve coding efficiency. Its simple and user-friendly interface makes it suitable for developing and testing the online shopping application.

4. DATABASE DESIGN

4.1 Database Tables

The Online Shopping Cart Simulation uses a relational database to store and manage user, product, cart, and order information in a structured manner. The database consists of four main tables: **User, Product, Cart, and Orders**.

- The **User table** stores details of registered users such as name, email, username, and password.
- The **Product table** stores information about available products including product ID, name, price, and quantity.
- The **Cart table** stores the items selected by the user, including product details and quantity.
- The **Orders table** stores order-related details such as customer information, payment method, and order date.

This separation of data helps in reducing redundancy and improves data organization.

4.2 Table Structure

4.2.1 User Table

Field Name	Data Type	Description	Constraint
User _ id	NUMBER	User ID	Primary Key
first _ name	VARCHAR2(50)	First Name	Not Null
last _ name	VARCHAR2(50)	Last Name	-
Email	VARCHAR2(100)	Email Address	Not Null
username	VARCHAR2(50)	Username	Unique
password	VARCHAR2(50)	Password	Not Null

Table 1: User Relational Database Table

4.2.2 Product Table

Field Name	Data Type	Description	Constraint
Product _ id	NUMBER	Product ID	Primary Key
Product_ name	VARCHAR2(100)	Product Name	Not Null
price	NUMBER	Product Price	Not Null
quantity	NUMBER	Available Stock	-

Table 2: Product Relational Database Table

4.2.3 Cart Table

Field Name	Data Type	Description	Constraint
Cart_ id	NUMBER	Cart ID	Primary Key
Product_ id	NUMBER	Product ID	Foreign Key
Product_ name	VARCHAR2(100)	Product Name	Not Null
Price	NUMBER	Product Price	Not Null
quantity	NUMBER	Quantity Selected	Not Null

Table 3: Cart Relational Database Table

4.2.4 Orders Table

Field Name	Data Type	Description	Constraint
order_ id	NUMBER	Cart ID	Primary Key
Product_ id	NUMBER	Product ID	Not Null
Product_ name	VARCHAR2(100)	Product Name	-
Price	NUMBER	Product Price	Not Null
quantity	NUMBER	Quantity Selected	Not Null
Cart_ id	NUMBER	Cart ID	-

Table 4: Orders Relational Database Table

4.3 Keys and Constraints

Keys and constraints are used to maintain data integrity and ensure accurate data storage.

- **Primary Keys:**
  - user_ id in User table
  - product_ id in Product table
  - cart_ id in Cart table
  - order_ id in Orders table
- **Foreign Key:**
  - product_ id in Cart table references product_ id in Product table
- **NOT NULL Constraint:**
  - Important fields such as product_ name, price, and quantity cannot be left empty
- **UNIQUE Constraint:**

    - Username in User table must be unique
- **Data Type Constraints:**
    - Each field accepts only valid data types to ensure correctness

These constraints help in maintaining consistency and avoiding duplicate or invalid data.

4.4 Table Relationships

The tables in the Online Shopping Cart Simulation are logically connected:

- A user can add multiple products to the cart
- Each cart contains multiple product entries
- A user can place multiple orders
- The cart acts as an intermediate stage before placing an order

This represents:

- One-to-many relationship between User
- Orders and One-to-many relationship between Product and Cart.

5. SYSTEM DESIGN

5.1 System Architecture

The Online Shopping Cart Simulation follows a **three-layer architecture**:

1. **Presentation Layer (User Interface)**
   - Developed using Java Swing
   - Allows users to interact with the system
2. **Application Layer (Business Logic)**
   - Processes user inputs
   - Performs operations like product selection, add to cart, update quantity, remove items, and billing calculation
3. **Data Layer (Database)**
   - Stores user, product, cart, and order data
   - Managed using a relational database such as Oracle or MySQL

This architecture ensures better organization, security, and maintainability.

5.2 Modules Description

The system is divided into the following modules:

1. **User Management Module**
   - Allows new users to register by providing required details
   - Provides login functionality for registered users
   - Validates user credentials for secure access
   - Manages user information such as username, email, and password
2. **Product Module**
   - Displays available products with details such as name, price, and quantity
   - Allows users to browse and select products
   - Maintains product information in the database
3. **Cart Module**
   - Searches employee by ID or name
   - Allows users to add products to the shopping cart
   - Updates quantity of selected items
   - Removes unwanted products from the cart
   - Displays cart items in a structured format
4. **Order Module**
   - Calculates total cost of selected products

- Generates final bill during checkout
- Stores order details such as customer information and payment method

5. **Database Connectivity Module**
   - Connects Java application with Oracle DB
   - Executes SQL queries
6. **User Interface Module**
   - Provides GUI using Swing
   - Handles user interactions

6. SYSTEM IMPLEMENTATION

Login Screen:

The Login Screen allows registered users to access the system by entering their username and password.

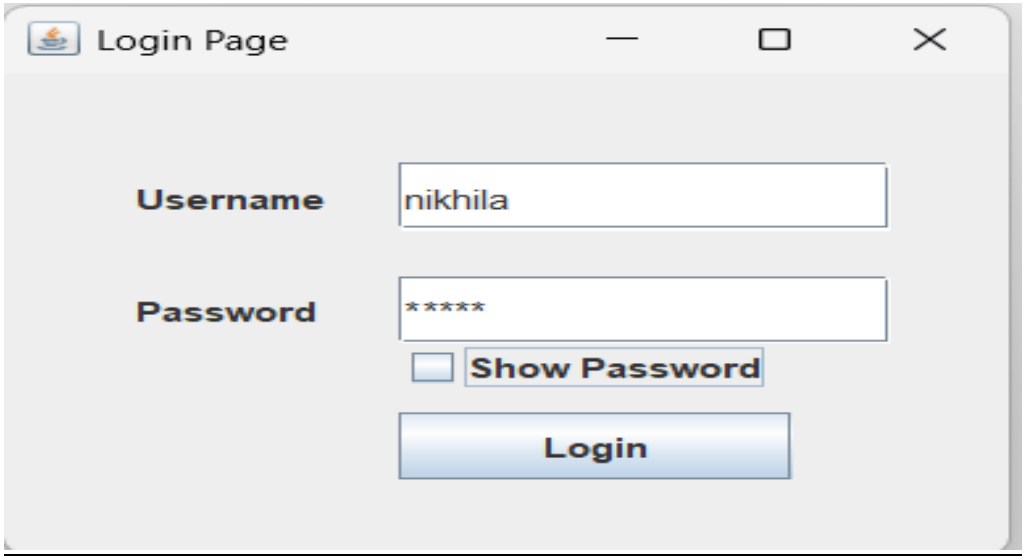


Figure 1: Login Screen of Online Shopping Cart Simulation

Main Menu Screen:

The Main Menu Screen provides options such as Login, Register, View Products, View Cart, Order History, and Logout.

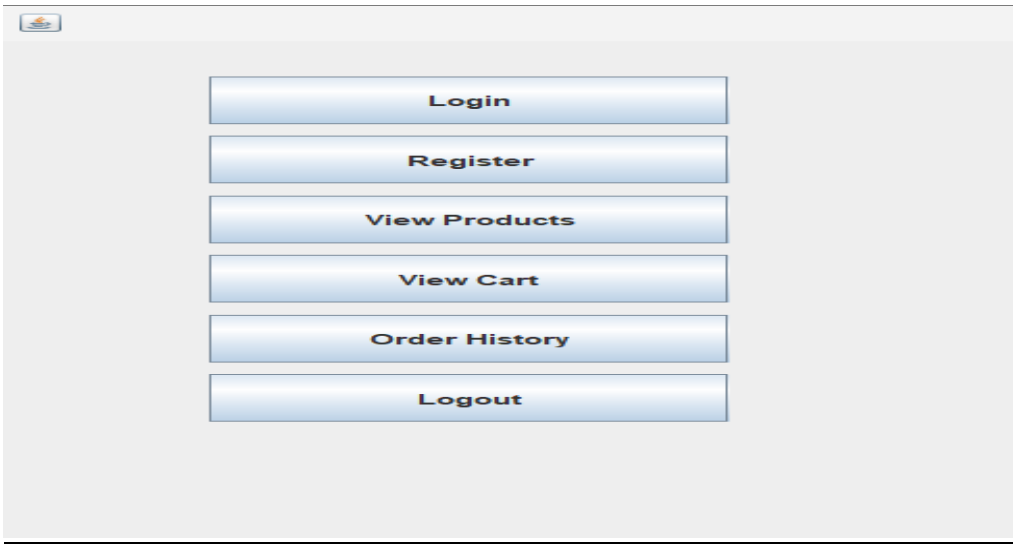


Figure 2: Main Menu Screen of Online Shopping Cart Simulation

Product List Screen:

Displays products in a structured list with options to view details, add items to the cart.

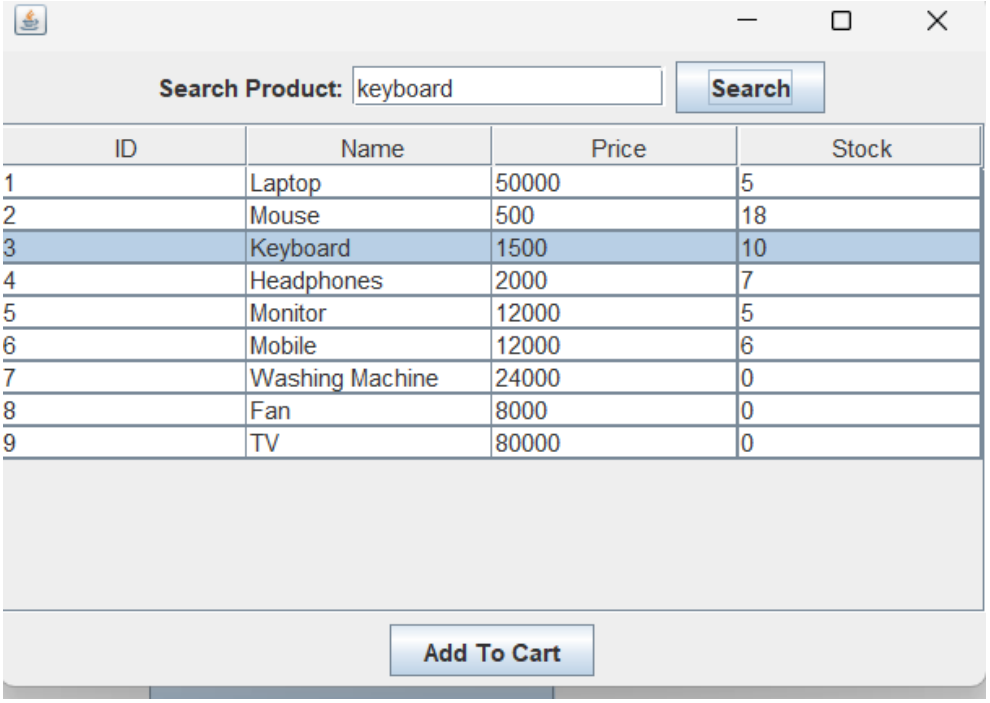


Figure 3: Product List Display Screen of Online Shopping Cart Simulation

Add to Cart Screen:

Allows users to add selected products to the cart and manage item quantities.

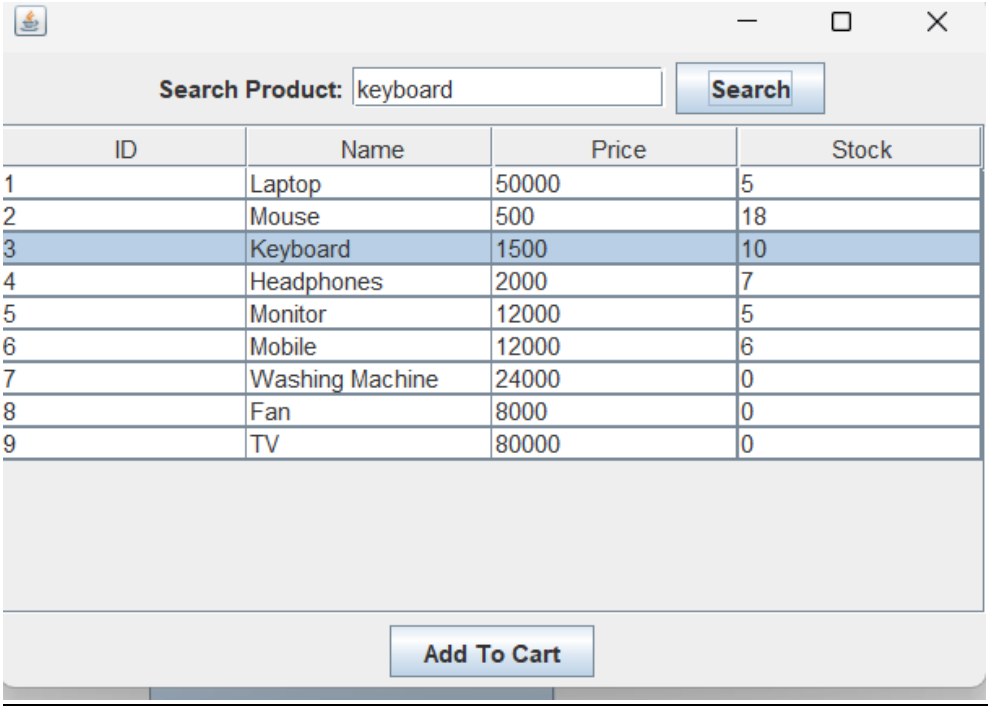


Figure 4: Add Product to cart Screen of Online Shopping Cart Simulation

View Cart Screen:

Shows cart items with options to update or remove products.

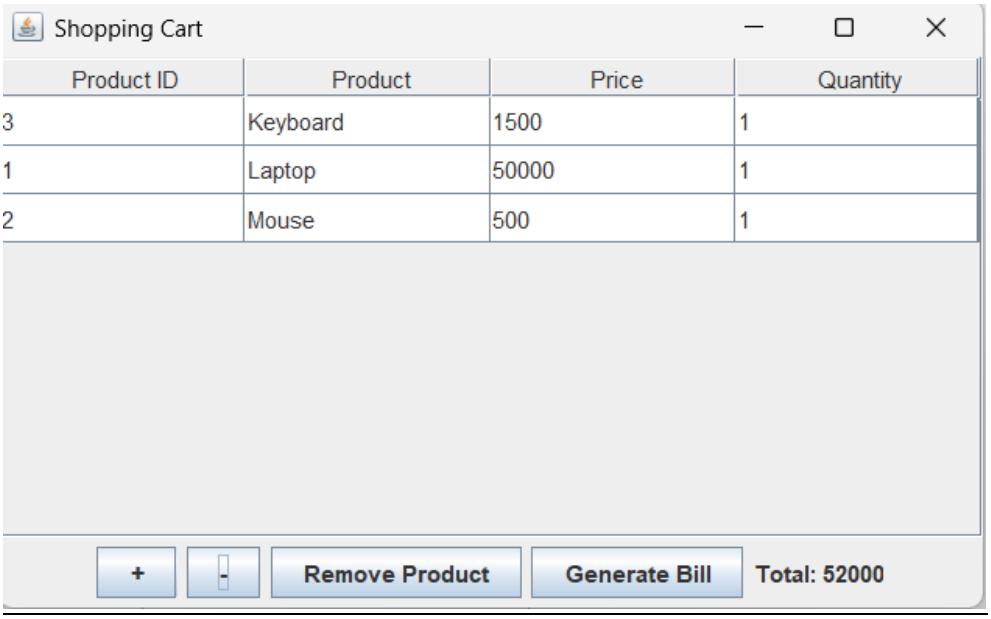


Figure 5: View Cart Screen of Online Shopping Cart Simulation

Update Cart Screen:

Allows users to modify item quantities or remove products in the cart.

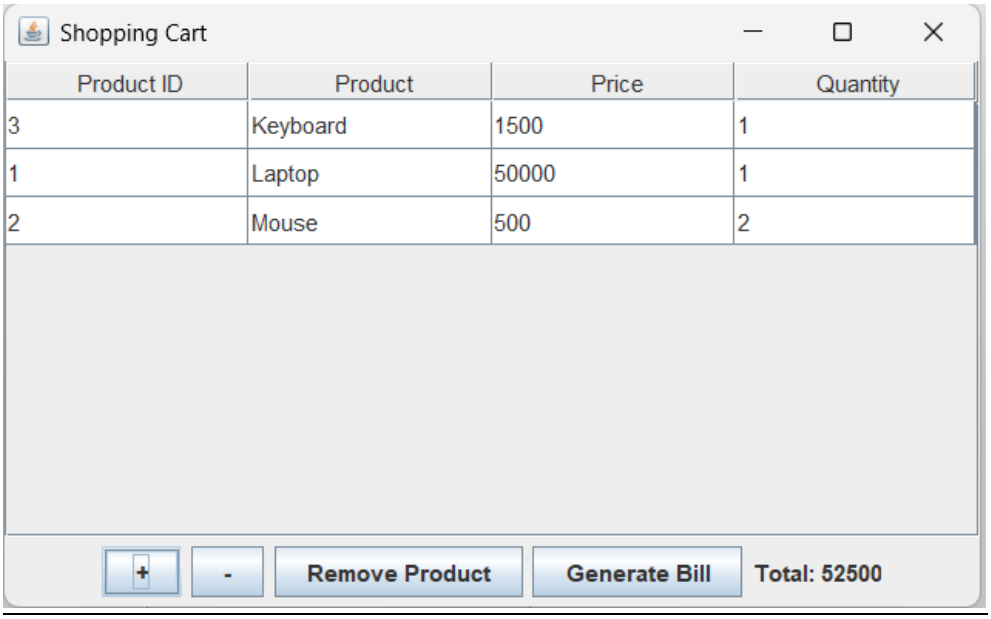


Figure 6: Updating/Remove item from Cart Information Screen of Online Shopping Cart Simulation

Bill Generation Screen:

Generates the final bill with item details, quantities, and total amount.

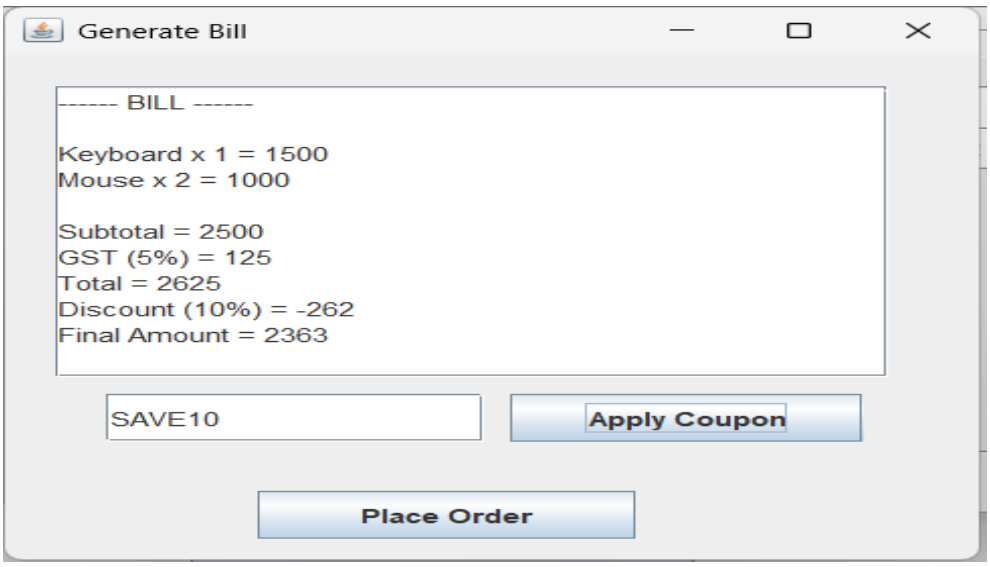


Figure 7: Bill Generation Screen of Online Shopping Cart Simulation

Order Confirmation Screen:

Confirms the order and displays a summary of purchased items and total amount.

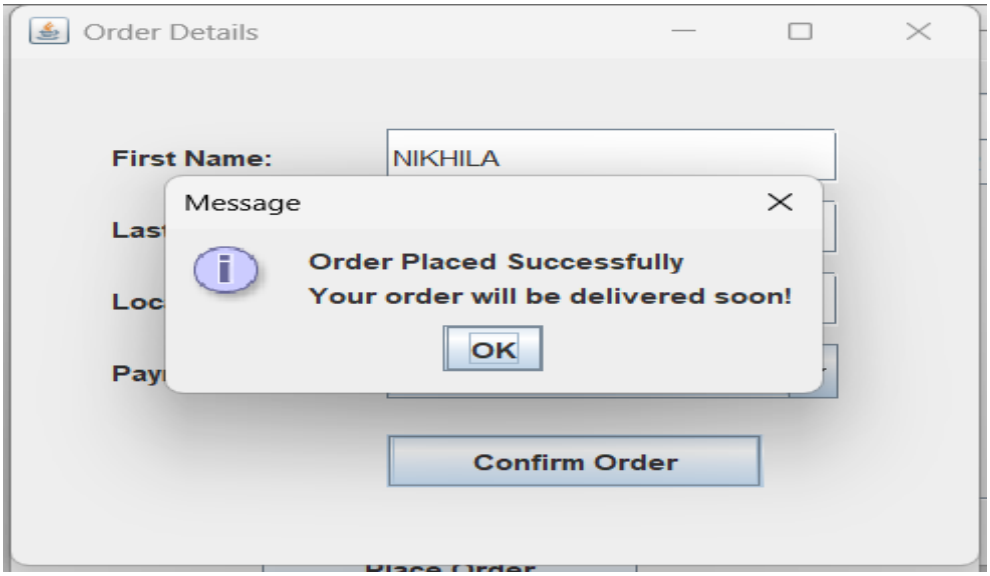
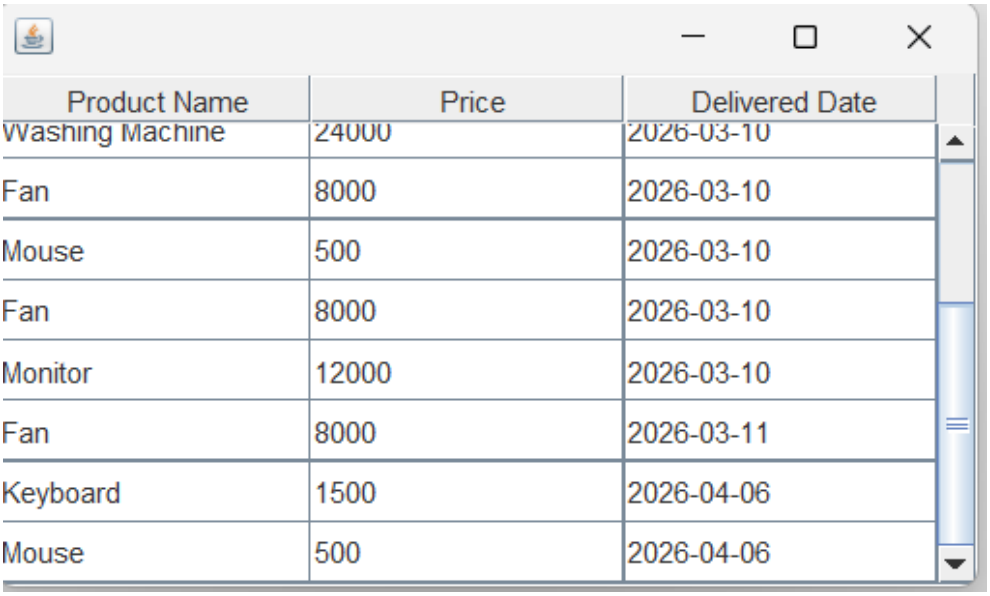


Figure 8: Order Confirmation Screen of Online Shopping Cart Simulation

Order History Screen:

Displays past orders with details and order status.



Product Name	Price	Delivered Date
Washing Machine	24000	2026-03-10
Fan	8000	2026-03-10
Mouse	500	2026-03-10
Fan	8000	2026-03-10
Monitor	12000	2026-03-10
Fan	8000	2026-03-11
Keyboard	1500	2026-04-06
Mouse	500	2026-04-06

Figure 9: Order History Screen of Online Shopping Cart Simulation

7. CONCLUSION

The Online Shopping Cart Simulation has been successfully developed to provide a smooth and efficient shopping experience for users. The system replaces traditional shopping methods with a digital solution, thereby saving time, reducing manual effort, and minimizing errors in order processing.

The application offers essential functionalities such as adding products to the cart, updating quantities, removing items, and calculating the total cost through a user-friendly interface. By using technologies like Java, Swing, JDBC, and Oracle Database, the system ensures reliable performance, secure transactions, and efficient data handling.

Overall, the system enhances the convenience of online shopping and simplifies the process for both users and administrators. It also provides a scalable solution that can be further improved by adding features such as online payment integration, order tracking, and user authentication.

In conclusion, the Online Shopping Cart Simulation meets the project objectives and serves as an effective tool for managing and simulating an e-commerce shopping experience.

8. FUTURE ENHANCEMENTS

The Online Shopping Cart Simulation developed in this project provides basic functionalities for managing online purchases. However, the system can be further enhanced by adding advanced features to improve its functionality and user experience.

- **Online Payment Integration:**
  Adding secure payment options such as credit/debit cards, UPI, and net
- **Web-Based Application:**
  Converting the desktop application into a web-based system for remote access
- **Order Tracking:**
  Providing real-time tracking of orders and delivery status
- **User Authentication:**
  Implementing secure login, registration, and role-based access
- **Improved User Interface:**
  Enhancing the design for better user experience

These enhancements can make the system more scalable, secure, and suitable for real-time organizational use.

9. REFERENCES

The following references were used during the development of the Employee Management System:

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8. **W3Schools Java Tutorial**,
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APPENDIX

A.1 Sample Code

Database Connection Code:

The following code establishes a connection between the Java application and the Oracle database using JDBC.

***DBConnection.java*:**

```
import java.sql.Connection;
import java.sql.DriverManager;
public class DBConnection {
public static Connection getConnection() {
Connection con = null;
try {
Class.forName("oracle.jdbc.driver.OracleDriver");
Con=DriverManager.getConnection("jdbc:oracle:thin:@localhost:1521:xe","syst
em","nikhila");   // your password
System.out.println("Database Connected");
} catch (Exception e) {
e.printStackTrace();
}
return con;   // return without closing
}
}
```

Dashboard code:

The dashboard code displays the main interface of the Online Shopping cart.

Dashboard.java:

```
import javax.swing.*;

public class Dashboard extends JFrame{

    Dashboard(){

        setLayout(null);
```

```
    JButton login = new JButton("Login");
    JButton register = new JButton("Register");
    JButton products = new JButton("View Products");
    JButton cart = new JButton("View Cart");
    JButton history = new JButton("Order History");
    JButton logout = new JButton("Logout");

    login.setBounds(80,30,200,40);
    register.setBounds(80,80,200,40);
    products.setBounds(80,130,200,40);
    cart.setBounds(80,180,200,40);
    history.setBounds(80,230,200,40);
    logout.setBounds(80,280,200,40);

    add(login);
    add(register);
    add(products);
    add(cart);
    add(history);
    add(logout);

    login.addActionListener(e -> new LoginPage());
    register.addActionListener(e -> new RegisterPage());
    products.addActionListener(e -> new ViewProducts());
    cart.addActionListener(e -> new CartPage());
    history.addActionListener(e -> new OrderHistory());
    logout.addActionListener(e -> {

int choice = JOptionPane.showConfirmDialog(
    this,
    "Are you sure you want to logout?",
    "Logout",
    JOptionPane.YES_NO_OPTION
);
```

```
            if(choice == JOptionPane.YES_OPTION){

                Session.isLoggedIn = false;

                JOptionPane.showMessageDialog(this,"Logged out successfully");

                dispose();
                new LoginPage();
            }

        });

                setSize(350,300);
                setVisible(true);
                setDefaultCloseOperation(EXIT_ON_CLOSE);
            }
        }
```

A.2 SQL Queries

Create User Table:

```
CREATE TABLE users (
  user_id NUMBER PRIMARY KEY,
  first_name VARCHAR2(50),
  last_name VARCHAR2(50),
  email VARCHAR2(100),
  username VARCHAR2(50),
  password VARCHAR2(50)
   );
```

Create product Table:

```
CREATE TABLE products (
   product_id NUMBER PRIMARY KEY,
   product_name VARCHAR2(100),
   price NUMBER,
```

```
    quantity NUMBER
);
```

Create Cart Table:

```
CREATE TABLE cart (
    cart_id NUMBER PRIMARY KEY,
    product_id NUMBER,
    product_name VARCHAR2(100),
    price NUMBER,
    quantity NUMBER
);
```

Create order Table:

```
CREATE TABLE orders (
    order_id NUMBER PRIMARY KEY,
    user_id NUMBER,
    product_id NUMBER,
    product_name VARCHAR2(100),
    price NUMBER,
    quantity NUMBER,
    total_amount NUMBER,
    order_date DATE
);
```